KEL452
Revised January 22, 2010

KENT GRAYSON AND MOHANBIR SAWHNEY

Teaching Note:
Ontela PicDeck (A) and (B)

Synopsis

Ontela is a technology start-up company that has introduced an innovative service called PicDeck that improves the mobile imaging experience for wireless subscribers. PicDeck allows wireless subscribers to easily and conveniently transfer pictures that they take on their mobile devices to their desktop computers, social networking sites such as Facebook, or any other networked devices. Ontela sells PicDeck to wireless carriers, who in turn private-label the service to their subscribers. Ontela must decide which customer segments it should target for the PicDeck service and how to create a positioning strategy and a marketing communication plan to promote its PicDeck service. It must also consider the value proposition of the PicDeck service for wireless carriers (its direct customers), who need to be convinced that the service will lead to higher monthly average revenue per user (ARPU) and/or increased subscriber loyalty.

The case is written in two parts. Part A provides qualitative information on customer personae that represent different customer segments, and students are asked to come up with a targeting and positioning strategy based on this qualitative information. Part B provides quantitative data on customer preferences that can be used to identify response-based customer segments, as well as demographic and media habits information that can be used to profile the segments. Students are asked to revise their recommendations on targeting and positioning based on the additional quantitative data. They are also asked to come up with recommendations on a media plan that can effectively reach their chosen target segments.

Teaching and Learning Objectives

This case has three main teaching purposes:

- To reinforce the principles of data-driven customer segmentation
- To discuss and emphasize the appropriate criteria for selecting segments
- To provide a deeper understanding of the benefits and drawbacks of different approaches to identifying and evaluating segments

MBA students and managers often say that they understand the principles of segmentation, but they are not sure how to design a quantitative customer segmentation study and use the results to come up with their targeting and positioning strategy. This case gives students a chance to roll up their sleeves and deal with the data produced by segmentation studies. It also illustrates how data-driven

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segmentation can produce results that may run counter to approaches that rely on "gut feel" or qualitative information alone.

Options for Teaching the Case

The case has two parts. Part A introduces the case and presents qualitative data that can be used to identify customer segments. Part B presents the segments that are identified using a cluster analysis of attitudinal and behavioral data. It also presents the demographic and media use data for the clusters identified through the cluster analysis.

The case can be taught in two ways.

Students can be given both parts together. The discussion can begin with the assignment questions in Part A, with students concentrating on the qualitative data on segmentation. The discussion can then proceed to the assignment questions in Part B to reinforce the value of adding attitudinal/behavioral data as well as demographic/media use data.

Alternatively, students can be given only Part A in advance and asked to address the assignment questions for Part A. Then, in a subsequent session (perhaps involving group work), students can be given Part B and asked to come up with their revised recommendations. The benefit of this sequential approach is that students cannot read ahead to find out what the additional data will reveal.

This case works very well as an assignment for teams of students, who work by themselves to come up with their recommendations. The class time can then be used to have a few teams present their recommendations, with the rest of the class acting as critics. Following these presentations, the instructor can debrief the case and highlight its key lessons. The assignment questions help structure the analysis, and students will benefit from having enough time to crunch the numbers and come up with well-thought-out recommendations.

Part 1: Introducing the Case

Why is Ontela even doing this research? Isn't it the wireless carriers who make decisions about which customers to target, which prices to charge, etc?

> Ontela can't make any money until it sells the service to wireless carriers. By developing recommendations about segmentation, targeting, and positioning (STP) for end customers, Ontela can more accurately predict the financial benefits that carriers will reap as a result of adopting the PicDeck technology. This knowledge will allow the firm to be more persuasive and successful when it interacts with its direct customers.
>
> From Ontela's perspective, the carrier is a customer, and that customer must be convinced that the cost of adopting, promoting, and servicing PicDeck will be more than offset by earnings from customer use of the technology.
>
> Furthermore, to the extent that carriers follow Ontela's advice about STP, and to the extent that this advice is good, Ontela can increase the chances that PicDeck will be successful in the marketplace.

Ontela's job is to identify valuable customer groups. What does it mean to be a "valuable" customer in this context?

In light of the previous discussion, it's useful to create two columns: "Value to Ontela" and "Value to Carriers."

Value to Ontela:	*Value to Carriers:*
The more a customer is willing to pay for a PicDeck subscription and/or the longer they hold the subscription, the more valuable s/he is. This is because Ontela earns a percentage of the subscription fees paid to carriers for the technology.	From the perspective of the wireless carrier, a customer is valuable for the same reasons as above, because carriers keep a percentage of the subscription fee. In addition, end-user adoption of PicDeck provides value to carriers to the extent that adoption results in greater wireless service revenues for the carrier. This might come from any of the following: • Subscribers who do not have a data plan may upgrade their service and sign up for a data plan. • Subscribers who have a data plan may use more data/photos and upgrade to a higher-level data plan. • Some subscribers may exceed the limits of their existing service package because of the intensive data requirements of pictures and end up paying a premium for sending data. • If PicDeck is not available from other carriers, subscribers may be motivated to switch from another carrier (because they want PicDeck) or it may foster greater loyalty among existing subscribers (because they don't want to lose PicDeck).

Of course, given the limited information we have about end-user behavior, these benefits are somewhat speculative. Given the switching costs in changing wireless carriers, it would be unusual for a customer to switch because of a relatively minor application like PicDeck. However, the instructor can note that PicDeck will be successful in its sales efforts to the extent that it can document and quantify the above advantages to carriers.

Part 2: How Useful Is a Customer Persona?

This section of the teaching note focuses on teaching Part A of the case.

LEARNING GOALS

- To discuss (or reinforce) the principle that attracting different customers requires developing different positioning statements (and thus implementing different marketing tactics).
- To illustrate that, when segmentation approaches are hypothetical and qualitative, it can be easy to develop a general strategy but difficult to develop a strong business case. Because decisions about hypothetical/prototypical segments tend to based on personal opinion and judgment, completely different conclusions can be reached using the same information.

DISCUSSION QUESTIONS AND ANSWERS

The case asked you to decide which segment Ontela should target. What criteria should companies use to decide which segment(s) to target?

One general criterion that tends to come up in discussion is how much consumers in the segment want/need the product or service. How can we evaluate whether a product like PicDeck will be wanted/needed by a particular customer?

One way to think about this is to consider (1) whether PicDeck has an attractive feature, and (2) if the feature provides an attractive and differentiated benefit. (An attractive benefit is one that addresses an unmet and important customer need.)

A second set of criteria tend to address the common question of how much value/money the segment is likely to generate. Financial value is a function of revenue minus costs, and is likely to be related to (1) how big the segment is, (2) customers' willingness to pay, and (3) the cost to identify, attract, and serve customers in the segment.

Attractiveness and value are related. For example, the more a customer values the service, the more easily the customer can be converted and the more financial value s/he is likely to provide.

Based on the feature/benefit analysis, what positioning statements are likely to be appropriate for each of the customers?

There are many ways to structure a positioning statement. At the very least, a statement should identify the customer, define the service or product, identify the benefit, and communicate why the benefit is differentiated from the competition. (See **Exhibit TN1** for ideas.)

One possible framework for a positioning statement is: For (target customer) who (statement of need/opportunity), PicDeck is (product/service category) that (statement of benefit). Unlike (primary competitive alternatives), PicDeck (statement of primary differentiation).

Developing positioning statements for this case demonstrates to students that the same feature can produce different benefits for different customers, necessitating different positioning statements and therefore different marketing activities.

What if a carrier wanted to position PicDeck for two of these segments? What would a positioning statement look like?

This leads to discussion about the challenges of targeting two different segments that value a product or service for different reasons and/or in different scenarios. The challenge is to find a way to communicate both benefits with the same proposition. An even more difficult challenge is to find a way to communicate two different positions in the market.

This challenge can sometimes be overcome with clever thinking, but not always. And in any case, the costs and benefits must be carefully considered. The obvious benefit of targeting two segments is that the number of potential customers becomes larger. However, the drawback is often that the positioning statement becomes more general, which may reduce the likelihood of conversion. Or, if the company chooses two different positioning

statements, it loses efficiency on the company side (two different ad campaigns) and may create confusion on the customer side.

Which segment should Ontela pick?

Having identified different criteria for picking segments, ask students to rate each segment relative to the other in terms of (1) how much customers in the segment are likely to want/need PicDeck, and (2) how much value customers in the segment are likely to provide in total to PicDeck and its partners.

After the analysis is done, pick a student and get his or her ranking and reasoning. Then ask for the opinion of someone whose ranking is quite different, and get his or her reasoning. Allow for some discussion/argument on both sides.

Then point out that this is the problem with personae that are based on assumptions rather than actual customer data: while it is often possible to develop a high-level marketing strategy (e.g., benefits and position) based only on personae, it is much more difficult to compare the attractiveness of different segments and/or develop specific marketing plans. We do not want marketing decisions to be based on who has the best argument (or who has the most seniority). Rather, we want decisions to be based on a true understanding of the customer. (It is sometimes also useful to point out that those who are arguing about what Regina or Sarah might do are not actually in the target segment.)

We come closer to a true understanding of customer preferences and motivations via the attitudinal data in the case.

Part 3: How Useful Is the Attitudinal and Product Space Information?

This section of the teaching note focuses on Part B of the case, with a particular focus on the attitudinal and product space information.

LEARNING GOALS

- How to systematically boil down a potentially overwhelming table of segmentation data
- The importance of understanding the difference between variables that distinguish customers from one another (but may not be important to our business) versus variables that are important to our business (but may not help distinguish customers from one another)
- The plusses and minuses of understanding clusters on their own versus relative to one another
- The value of approaching the data from different perspectives and looking for converging evidence

TEACHING PLAN FOR THIS PORTION OF THE DISCUSSION

Teaching this portion of the case can be done by systematically focusing on the data as follows:

1. Understanding and evaluating the variables. Which variables are more (or less) important for Ontela and its partners?

2. Assessing the relative financial value of each cluster. Which cluster is more (or less) likely to produce high revenue for Ontela and its partners?

3. Between-cluster and within-cluster analysis. Between-cluster analysis involves comparing how customers in *different* clusters rated the *same* questions. Within-cluster analysis involves comparing how a customer in a particular cluster rated *different* questions.

4. Holistic evaluation of the clusters. Having taken each of the above steps, students are in a better position to understand the relative attractiveness of each cluster and can come to a decision based on the insights provided by each

Much of this part of the case discussion focuses on a systematic analysis of the data in the exhibits. It can be helpful to project the data in spreadsheet form on the class screen. It can also be helpful to have a different worksheet for each analytic approach, perhaps with colors that highlight the important numbers for each approach.

Also, at each stage of the analysis it can be useful to keep track (e.g., on a flip chart) of which segments emerge as attractive from different perspectives. Generally speaking, if a segment seems valuable regardless of how the information is analyzed, this more strongly supports the value of a segment than if it seems valuable only when taking a particular perspective.

INTRODUCTION TO THE UNDERLYING STATISTICAL ANALYSIS

Depending on the case-study participants, some or all may not be familiar with the statistical analysis of customer data. It can therefore be helpful to explain the following:

- Standard statistical methods were used to group or "cluster" respondents together based on how similar their responses were.
- This analysis considers similarities in response profiles across all questions (not focusing merely on individual questions).
- This analysis does not presume that there will be a particular number of clusters. The statistical analysis guides the analyst to determine the number of clusters that best represent the survey respondents.

This analysis did not consider any information beyond the data provided by the questions. This means that demographic information was not considered.

ASSESSING THE RELATIVE FINANCIAL VALUE OF THE CLUSTERS

The survey results provide information about the subscription fee that each customer is likely to pay for the service. This number communicates two things at once: the value of the service to the customer and the customer's price sensitivity. (We can't separate the individual influence of each of these factors.)

The survey results also provide information about the likely size of each cluster as a percentage of the population. Different carriers service different areas and have customer bases of different sizes, so calculating total revenues for any carrier would require knowing how many potential or existing customers it has.

However, assuming that carriers tend to reflect the national average, the relative value of each customer segment can be calculated by multiplying the likely subscription fee by the percentage of total respondents represented by that cluster:

	Cluster					
	1	2	3	4	5	6
Monthly Fee	$5.75	$6.64	$7.93	$3.90	$8.31	$1.67
Percentage of Respondents in Cluster	24%	12%	19%	16%	15%	14%
Relative Value of Customer Segment (Fee × Percentage)	$1.38	$0.80	$1.51	$0.62	$1.25	$0.23

Relatively speaking, Cluster 3 is most valuable. Even though it is smaller than Cluster 1, the willingness to pay is higher. Interestingly, the third largest cluster, 4, is the second-least valuable because of its very low willingness to pay.

Potential Drawbacks of this Analysis. Survey respondents were given a generic description of PicDeck and its features. However, when PicDeck is launched, its features will be communicated in a way that emphasizes the particular value of PicDeck to the particular segment. If STP is done well, it should increase willingness to pay relative to WTP in response to a generic description. Nonetheless, this increase will likely occur for every segment that is interested in PicDeck (it is hard to develop a position that will increase WTP for customers not interested). Thus, the relative value of each attractive segment will probably not change much.

Another drawback is that we have not considered cost to attract and serve. It could be that the cost to attract and serve Cluster 3 is much higher than for Cluster 1, which might switch the relative value of the two clusters.

Based on this analysis, Clusters 1, 3, 5 are most attractive and Clusters 2, 4, 6 are least attractive.

UNDERSTANDING AND EVALUATING THE VARIABLES

Based on your experience and/or intuition, where do survey questions like these tend to come from?

> In a best-case scenario, the questions are developed based on initial exploratory customer research (e.g., interviews), which helps the analyst pick the best questions for understanding the customer.
>
> However, questions are also selected for a number of other reasons. Stakeholders other than marketing (product development, operations) sometimes vie for space on a survey. Market research firms sometimes have their own favorite questions. And firms sometimes adopt questions from previous surveys simply because "that's the way we've always surveyed customers."
>
> Because managers often receive market research studies without much explanation for the origin of the questions, it is useful to start by asking which questions are more or less important to the task at hand.

Which preference/attitudinal variables in Part B Exhibit 1 are most useful for segmentation identification and evaluation? Which variables are least useful?

- Questions 1–5 directly or indirectly address customer frustrations (sources of dissatisfaction).
- Questions 6–12 address the customer's behaviors and preferences related to the product space.
- Questions 13–14 allow assessment of segment attractiveness.

1. It is easy to upload phone photos
2. Would take more phone photos if transfer were easier
3. Would take more phone photos if quality were better
4. Would replace digital camera if phone transfer were easier
5. Would replace digital camera if phone photo quality were better
6. I value saving phone photos so they are not lost
7. I value sending photos by email
8. I value having photos on my desktop
9. I value sharing photos on Facebook, etc.
10. When deciding on new tech, easy setup is a factor
11. When new technologies emerge, I am first to adopt
12. When buying new tech, I search for best price
13. Monthly fee
14. Percentage of respondents in each cluster

To assess a segment's likely strength of interest in PicDeck, a marketing analyst needs to know the answer to two general questions. First, how much does the consumer use and value the photo content captured by a mobile telephone device? Q6–Q9 help to estimate the answer to this question. (Importantly, the analyst must be confident that these four questions address all the major ways of using photo content captured by a mobile telephone.)

Second, to what extent might the consumer's behavior change if transfer were easier? Three of the survey questions are of maximum importance in determining this:

- Q1 "It is easy to upload photos." This is the customer frustration that PicDeck is meant to address. If someone gives a rating above 3 on this, they will not value the technology as much as someone who is below 3—regardless of their answers to the other questions.
- Q2 "Would take more phone photos if transfer were easier." Because the main benefit of PicDeck is easy transfer, this question provides information on how PicDeck adoption might affect revenues for carriers (taking more phone photos for transfer means more wireless use).
- Q4 "Would replace digital camera if transfer were easier." Again, because the main benefit of PicDeck is easy transfer, this question provides information on how adoption might encourage a customer to use the camera phone more, transfer photos more, and therefore increase wireless use.

Because we are told that PicDeck is easy to set up, one further potentially useful question is Q10 "When deciding on new technology, easy setup is a factor." However, like all the questions, this one is useful only to the extent that the customer values the PicDeck offering in the first place.

It is important to understand how consumers' answers to Q6–Q9 might relate to the answers to Q1, Q2, and Q4. For example, consider Consumer A, who finds photo transfer difficult (Q1) but does not value having or posting pictures (Q6–Q9). Now consider Consumer B, who finds photo transfer difficult (Q1) but also values sharing photos via Facebook. Consumer B is likely more valuable to Ontela and its partners than Consumer A because PicDeck adoption is likely to result in more photo transfer for B, who values that advantage more than A does. Even if A also says that s/he will take more photos if transfer were easier (Q4), an increase in transfer ease may encourage A to change from transferring zero photos to three or four, while a similar increase may encourage B to change from transferring one dozen photos to two dozen.

There is a further potential dependency. A consumer may give low ratings to Q6–Q9 in part because s/he finds transfer difficult. That is, if transfer were easier, this consumer might find greater value in photos captured via cellphone. Although further research would likely be needed to identify these particular consumers, the survey nonetheless does a reasonable job identifying those who already value photos captured via cellphone and whose behavior would change if phone transfer were easier.

BETWEEN-CUSTOMER AND WITHIN-CUSTOMER ANALYSIS

To further synthesize the data, we can understand the ratings in two complementary ways. First, we can isolate each question and see which customer gave the question the rating that is most favorable to Ontela. This is a between-customer analysis, where the analyst focuses horizontally (rows). Second, we can isolate each cluster and see which attributes customers in each cluster are most enthusiastic about. This is a within-customer analysis, where the analyst focuses vertically (columns).

One approach to take for this part of the analysis is to ask half the class to focus horizontally and the other half to focus vertically. Sometimes even those who have sufficiently prepared for the case need five or ten minutes to orient themselves to one perspective or the other.

Students can become confused with the ratings for Q1 and Q12. For all the other ratings, a higher number tends to be better. For example, Ontela wants customers who strongly agree with Q2 "Would take more phone photos if transfer were easier." However, for Q1 and Q12, Ontela wants a lower number—those who do not currently think photo transfer is easy and/or those who do not shop on price. This can lead to a useful discussion about the importance of reading segmentation studies carefully and about the level of detail necessary to analyze these studies accurately. However, depending on the makeup of the class, instructors may want to avoid this discussion (and confusion) entirely by reversing the measures for discussion.

Between-Customer Analysis (Horizontal)

Regarding nearly all the items, customers in Cluster 1 tend to be the ones who have the strongest views. Customers in this cluster appear to be very involved with technology and are interested in greater involvement as well as in greater convergence between phone and camera. In contrast, those in Cluster 6 tend to give the lowest ratings.

There are a few exceptions. In terms of the highest ratings, those in Cluster 5 are least likely to agree that it is easy to upload photos (Q1), and those in Cluster 3 consider themselves most likely to adopt new technologies (Q11). In terms of the lowest ratings, those in Cluster 4 are most likely to agree that transferring photos is easy; those in Cluster 3 are least likely to agree that easy setup is a factor; and those in Cluster 2 are least likely to agree that they are early to adopt.

But generally speaking, this approach to understanding the clusters helps to clearly distinguish Clusters 1 and 6 from the others, but does little to help distinguish the remaining five clusters from one another. At this point, it can be helpful to ask students what they might suggest for further analyzing the data.

There are a number of useful options. For example, the analyst could begin by looking at consumers who gave the second-highest rating to each question. Or the analyst could consider consumers who gave a rating higher than the midpoint of 3. A third alternative is to look at all consumers who gave a higher than average rating. An analyst might reach the following important potential insights by using this third alternative:

- Cluster 3 is relatively unlikely to agree that uploading photos is easy and very likely to agree that they would take more photos if uploading were easier (these were two very important questions). Importantly, consumers in this cluster also place relatively high value on digital photos (Q6–Q9).
- Although Cluster 5 is also relatively unlikely to agree that uploading photos is easy, consumers in this cluster place a very low value on digital photos (Q6–Q9). These consumers might well become more interested in digital photos if they were easier to transfer, but the value of this cluster seems less dependable than the potential value of Cluster 3.
- Cluster 2 is very likely to agree that easy setup is important, which is a benefit of PicDeck, but not a central benefit. However, consumers in this cluster also value digital photos (Q6–Q9).

Potential Drawbacks of this Analysis. It assumes that customers both perceived the questions and used the rating scales in a comparable way. For example, consider the important item "It is easy to upload phone photos." Cluster 1, which is very technologically involved, gave this question roughly the same rating as Cluster 6, which is very technologically uninvolved (2.93 vs. 3.00). Although more research would be necessary to truly understand whether they are comparable, some potential concerns are that "easy" means different things to each of these consumers, and that respondents in Cluster 6 may not have had much experience with uploading photos, so their neutral rating might reflect their uncertainty about whether it is hard or easy.

A second drawback is that the cutoffs such as "above average" can create artificial distinctions. For example, using this approach, consumers in Cluster 5 are considered relatively likely to agree that easy setup is a factor (rating = 3.37) because they are above average, while those in Cluster 4 are considered relatively unlikely to agree with this statement (rating = 3.27) because they are below average.

Perhaps the most important drawback is that just because a customer in a particular cluster gave a higher rating than someone in another cluster does not mean that this item is most important *to him or her*. For example, relative to those in other clusters, those in Cluster 1 were most likely to agree that easy setup is a factor when adopting new technology. However, relative to Cluster 1's

responses to other items, this item was of least importance. This leads us to further analyze within-cluster ratings (rather than across clusters).

Based on this analysis, Clusters 1, 2, and 3 are most attractive, and Clusters 4 and 6 are least attractive.

Within-Customer Analysis (Vertical)

A within-customer analysis involves ranking which items are most important within each cluster (and/or which aspects each cluster is most enthusiastic about).

This kind of analysis produces a couple of interesting insights that are not very relevant to Ontela. These are as follows:

- The number one or number two item for all clusters was "Would take more phone photos if camera phone quality were better." This is an interesting insight but it does not distinguish clusters from one another, and it is not extremely relevant to Ontela, which has no control over phone camera quality.
- For Clusters 1 and 4, "Would replace digital camera if phone quality were better" was a top concern. But again, it is not directly relevant to PicDeck.

At this point it can be helpful to discuss what should be done when clear insights are generated that are not directly relevant to the project at hand. This is likely to be highly valuable information about customers that the company should consider capitalizing on via another product. At a customer-focused organization, potentially powerful insights like these would not be simply cast aside as irrelevant. There would be a well-known and easy channel for these insights to be communicated to those in charge of keeping track of new customer developments and insights.

The within-cluster analysis also produces findings that are more relevant to PicDeck:

- "It is easy to upload photos." To facilitate comparison with the other ratings, it can be helpful to reverse-code these ratings (by subtracting them from 5), thus indicating the extent to which they believed uploading photos was hard (and thus higher is better). Doing this highlights the fact that most customers cared more passionately about other issues than ease of uploading.
- "Would take more photos if transfer were easier." For Clusters 3 and 5, this was the second-highest item. This is important because easy transfer is a main selling point of PicDeck. (And note: this item received a very low rank for Cluster 4.)
- "Having photos on desktop" was a top concern for Cluster 2.
- "Easy setup" (an important potential attribute for Ontela) was valued by Cluster 6, but this cluster did not necessarily value other aspects of PicDeck's offering.

Potential Drawbacks of this Analysis. It may not be completely valid to compare responses to questions that address qualitatively different issues. For example, it is probably helpful to compare a customer's rating of "Would take more photos if transfer were easier" versus "Would take more photos if quality were better." However, it is probably less helpful to compare a customer's rating of "Would take more photos if transfer were easier" versus "When buying new technology, I tend to search for best price." However, this is a relatively minor drawback.

Based on this analysis, Clusters 2, 3, and 5 are most attractive.

UNDERSTANDING THE CLUSTERS MORE HOLISTICALLY

Clusters 3 and 5 emerge as attractive across multiple perspectives. Having begun prioritizing the segments, students should take a step back and try to understand these clusters holistically. What kind of customers are they? Do any of them resemble Sarah, Steve, or Regina? The list below is quite speculative, and students often have different opinions, but the list gives a flavor for the kind of discussion this can generate.

- Cluster 1: Price-sensitive tech enthusiast. Could be Regina?
- Cluster 2: Would not replace camera but values photos on desktop. Could be Sarah?
- Cluster 3: Wants photo convergence on phone, doesn't care about setup. Could be Regina?
- Cluster 4: Already finds transfer easy but doesn't like phone photo quality. Could be Regina?
- Cluster 5: Would take more photos if quality and transfer were easier; does not have photos on desktop and is a slow adopter. Could be Steve?
- Cluster 6: Technologically inexperienced and uninterested.

Also, note that five clusters have been identified but only three personae were initially developed. Did the segmentation study identify a potentially attractive segment that was not initially identified?

DEVELOPING A POSITIONING STRATEGY

At this stage, it is useful to ask different students to define the positioning strategy for different segments. Ask one student for his or her positioning statement for Cluster 3, and then ask another student the same question. Quite frequently, these positioning statements will be relatively similar. This emphasizes the fact that when you have a clearer idea of what motivates the customer, there is less disagreement about how to position the product.

At the end of this discussion, it is sometimes useful to emphasize that this attitudinal information was helpful in developing a defensible argument for selecting a segment, but it is less helpful in guiding the development of a marketing plan. We may know that a cluster is attracted to PicDeck but we cannot act on this information because we are unable to *reach* our target customers without more information on what they look like and what media they use. This is where the *profiling* information (demographics and media habits) becomes crucial. This information is provided in Part B Exhibits 3 and 4.

Part 4: How Useful Is the Demographic and Media Use Information?

This section of the teaching note focuses on Part B of the case, with a particular focus on the demographic and media use information.

LEARNING GOALS

- Attitudinal data tends to be better at revealing why consumers buy, but demographic information tends to be better at revealing who is buying.
- Segmentation data is useful only to the extent that it distinguishes customer groups from one another.
- Data-based market research can reveal customer groups that might not have been identified by intuition and educated guess alone.

INTRODUCTION TO THE UNDERLYING STATISTICAL ANALYSIS

Depending on how experienced students are with market research, it can be important to emphasize that the cluster analysis was based on attitudinal and preference data only. Demographics were subsequently added to help describe each of the clusters.

Cluster analysis can be done on the survey data only, on the demographic data only, or on the survey and demographic data combined. If students are relatively knowledgeable about market research techniques, it can be valuable to discuss the plusses and minuses of each approach.

DISCUSSION QUESTIONS AND ANSWERS

What variables most strongly distinguish the clusters from one another? Which are weakest?

Strong distinguishers:

- For the demographic characteristics, the most variance occurs when asking whether respondents have data plans, whether they have family plans, and whether they are married.
- For the media use information, the most variance occurs when asking respondents about technical magazines, celebrity magazines, and Internet use.

Weak distinguishers:

- For the demographic characteristics, gender has the least variance.
- For the media use characteristics, women's magazines have the least variance (respondents tended not to read these).

To what extent does the demographic and media use data line up with our earlier analysis?

If everyone has done this analysis, you can divide the room into six groups, asking each to evaluate one of the individual clusters based on the educated guesses you previously discussed. For example, to what extent does Cluster 1 seem like Regina? To what extent does Cluster 2 seem like Sarah?

For discussion purposes, it can be useful to end with an analysis of Cluster 1 and 4.

Cluster 2. If we recognize that "Sarah" could be male or female, this cluster seems convergent with our idea of this persona. Of all clusters, this one is most likely to be married and among the most likely to have a family plan. This cluster is also least likely to have a data plan (Sarah is technologically lagging). Customers in this cluster also show

stronger interest in fashion and homemaking magazines. (Remember that all clusters have a lot of men in them, so a rating of 2.65 for fashion magazines is potentially revealing.)

Cluster 3. If we recognize that "Regina" could be male or female, this cluster seems convergent with our idea of this persona. This is the youngest cluster and the least likely to be married. It is also one of the most likely clusters to have a message pack (lots of texting) and to have a family plan. This cluster also has the highest ratings for celebrity, fashion, and women's magazines, which also fits this persona.

Cluster 5. Steve could be this cluster, but the demographics and media use statistics are all in between the extremes. There are some tendencies—a little older than average, a little more likely to be married than average, a little less likely to have a message pack or data plan. They are less likely than average to read a fashion magazine. This does not go against what we know about Steve but the numbers are potentially too lukewarm to really use them as identifiers. It could be that this cluster is too demographically diverse to "nail" using demographics.

Cluster 6. Gives us a better picture of the technologically uninterested customer. This is by far the oldest cluster and among the most likely to be married. This cluster is also among the least likely to have a family plan, message pack, or data plan.

These clusters seem to map reasonably well onto the personas from the first part of the case. However, two other clusters seem to have characteristics that make them unique.

Cluster 1. As a tech-focused cluster, Regina is the only possible candidate. But even if we recognize that "Regina" could be male or female, this cluster seems unlikely to be this persona. Cluster 1 is about average in age (not younger) and the percentage married is average (not lower than average). Also, there is relatively high reading of sports and tech magazines, which does not fit the Regina profile.

Cluster 4. Regina is less likely to be this cluster. It is on the younger side but not as young as Cluster 3. It is the most strongly male cluster. Among the clusters, this one is most likely to have a data plan and least likely to have a message pack. This cluster also heavily reads tech and sports magazines.

Thus, this analysis has revealed two potential customer groups that were not identified in the persona analysis. It is possible that they need to better understood via further inquiry.

Based on our analysis so far, what are the benefits and drawbacks of doing persona development before quantitative cluster analysis versus cluster analysis before persona development?

The advantage of doing persona development first is that it can guide the questions you choose for the survey you will use to collect data for cluster analysis.

The advantage of doing persona development second is that the personae are based on hard market data, which is more likely to reflect reality than intuition or hunch.

Because both approaches have their advantages, market research is best done iteratively: Develop initial personas first, then adjust based on market research data.

Which clusters would you choose based only on the demographic data?

If based on the demographic data only, we might lean toward:

- Those with high Internet use (Clusters 1, 3, 4)
- Those more likely to have a message pack (Clusters 3, 4)
- Those more likely to have a data plan (Cluster 1)

Building on the personae, it may be possible to identify Cluster 3 as Regina and Cluster 2 as Sarah.

However, while the demographic data is somewhat helpful in identifying *who* to target, it is not very helpful in identifying *why* we should target them. It may be based partly on current usage but not current need. For example, neither Sarah nor Steve has high Internet use, but they might be attractive customers just for that reason. And it turns out that Cluster 4 already finds it easy to transfer pictures from the phone, so even though they are more likely to have a message pack they are not attractive to Ontela/PicDeck.

Exhibit TN1: Table for Discussing Customer Personae

		Sarah	Steve	Regina
How much do customers in the segment want/need the product/service?		After discussion, rate Sarah relative to the others on a scale from 1 to 5	After discussion, rate Steve relative to the others on a scale from 1 to 5	After discussion, rate Regina relative to the others on a scale from 1 to 5
	Most attractive Ontela feature	Makes it easier to transfer photos from phone to computer	Makes it easier to transfer photos from phone to clients	Makes it easier to transfer photos from phone to social sites
	Most attractive Ontela benefit	Save memories	Improve business performance	Stay better connected/ be more popular
Value to Ontela and partners		After discussion, rate Sarah relative to the others on a scale from 1 to 5	After discussion, rate Steve relative to the others on a scale from 1 to 5	After discussion, rate Regina relative to the others on a scale from 1 to 5
	Segment size	It is possible to develop a rough view of relative size. For example, if Regina can be male or female, this segment is probably larger than Sarah or Steve. That helps with relative attractiveness but not with developing a strong business case.		
	Willingness to pay	WTP is more speculative. Will Regina herself pay for the service, or will her parents? If Steve can deduct as a business expense, how much will his WTP go up? Are Sarah and Regina in the same family?		
	Cost to identify, attract, and serve	Demographically, these customers are differentiated and relatively easy to identify. The rest is more speculative. In terms of cost to attract, Regina is probably the most media savvy, so marketing communications must be more creative. Steve might be most demanding (higher cost to serve) because he is using the service for his business.		
Positioning statement		For moms who want to preserve memories, PicDeck is a simple cellphone feature that easily and automatically transfers phone photos to your desktop computer. Now your cellphone photos won't accumulate for months on your telephone!	For professionals whose clients demand immediate photographs, PicDeck is a simple cellphone feature that easily and automatically sends phone photos to your clients' email accounts. Rather than waiting to send photos when you get back to the office, PicDeck will allow you to serve your clients more quickly and effectively.	For popular teens who want to stay connected with their friends, PicDeck is an advanced cellphone feature that automatically posts phone photos on your Facebook page. Rather than waiting to post photos when you get home, PicDeck will let you post photos in real time, as it happens!